

Last time Week 2-2

For Bernoulli $\text{Ber}(p)$ MLE $\hat{p} = \bar{x}$

For Normal $N(\theta_1, \theta_2)$ MLE $\hat{\theta}_1 = \bar{x}$

$$\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Note: ~~Estimates~~ $\hat{\theta}_1, \hat{\theta}_2, \hat{p}$ are random variables
(approximating μ, σ^2, p)

$$\rightarrow E\hat{p} = E\bar{x} = EX_1 = p$$

$$\rightarrow E\hat{\theta}_1 = E\bar{x} = EX_1 = \mu = \theta_1$$

other way:

$$X \sim N(\mu, \sigma^2) \Rightarrow \bar{X} = \frac{\sum X_i}{n} \sim N\left(\frac{n\mu}{n}, \frac{\sigma^2}{n}\right) \text{ is normal too}$$

$$E\bar{X} = \frac{n\mu}{n} = \mu$$

$$\rightarrow E\hat{\theta}_2 = E\left(\frac{1}{n} \sum (x_i - \bar{x})^2\right) = \frac{1}{n} E\sum (x_i - \bar{x})^2 \neq E\sum (x_i - EX)^2$$

since $\bar{x} \approx EX$
but not $= EX$

other way?

find distribution of $\sum_{i=1}^n (x_i - \bar{x})^2$ (§ Section 5.5 content)

$$\text{if } X \sim N(\mu, \sigma^2), \text{ then } Z := \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$V := \frac{(X - \mu)^2}{\sigma^2} = Z^2 \sim \chi^2(1)$$

Chi-square distribution with one degree of freedom

$$\chi^2(r): f(x) = \frac{x^{r/2-1} e^{-x/2}}{\Gamma(r/2) \cdot 2^{r/2}} \quad \text{PDF} \quad x \in (0, \infty)$$

$$EX = r \quad \text{var } X = 2r \quad \text{MGF} = \frac{1}{(1-2t)^{r/2}}$$

PDF: $\overset{\text{CDF}}{F}(t) = P(Z^2 \leq t) = P(-\sqrt{t} \leq Z \leq \sqrt{t}) =$

$$= 2 \int_0^{\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

change of var
 $y = x^2$

$$= 2 \int_0^t \frac{1}{\sqrt{2\pi} \sqrt{y}} e^{-y/2} dy$$

Thm of calculus

PDF $f(t) = F'(t) =$

$$\frac{1}{\sqrt{2\pi y}} e^{-y/2} = \frac{e^{-y/2} y^{-1/2}}{2^{1/2} \Gamma(1/2)}$$

$\Gamma(n) = n!$ if n - integer

⊗ $\Gamma(s) = \int_0^\infty e^{-x} x^{s-1} dx$ for $s > 0$ normalization constant

$\Gamma(\frac{1}{2}) = \sqrt{\pi}$

MGF $M(t) = E(e^{tx})$

Exercise 3.2-7
Technique from section 3.1

Facts: $X_1 + \dots + X_n \sim \chi^2(r_1 + \dots + r_n)$ if $X_i \sim \chi^2(r_i)$

• in particular $X_i \sim \chi^2(1) \Rightarrow \sum X_i \sim \chi^2(n)$

(MGF($\sum X_i$) = MGF($\chi^2(n)$)) - check this!

• $[\sum (X_i - \bar{X})^2] \overset{\text{Thm 5.5-2}}{\sim} \chi^2(n-1)$

(Pf) $\chi^2(n) \sim \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} = \sum_i \frac{[(X_i - \bar{X}) + (\bar{X} - \mu)]^2}{\sigma^2}$

$= \frac{\sum (X_i - \bar{X})^2}{\sigma^2} + \frac{\sum (\bar{X} - \mu)^2}{\sigma^2} + \text{cross terms}$ (check this!)

$$T^2 = \left[\frac{n(\bar{X} - \mu)}{\sigma} \right]^2 \sim \chi^2(1) \quad \text{since} \quad T = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

since $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$\frac{\sum (X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n) - \chi^2(1) \sim \chi^2(n-1)$$

$$E\hat{\theta}_2 = E(X_i - \bar{X})^2 = \sigma^2 \cdot E(\chi^2(n-1)) = \sigma^2(n-1) \neq \sigma^2$$

$$E\left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right) = \sigma^2$$

$\hat{s}^2$ (sample variance)

Def If $E(u(X_1, \dots, X_n)) = \theta$, then $u(X_1, \dots, X_n)$ is unbiased estimator
 Otherwise it is biased.

- $\hat{\mu}, \hat{\theta}_1$ are unbiased
- $\hat{\theta}_2$ is biased for σ^2
- $\hat{s}^2$ is unbiased for σ^2

Method of moments

if $L(\theta)$ is hard to optimize

empirical moments $E(X^k) \approx$ moments of PDF

Ex) $N(\theta_1, \theta_2)$

$$EX = \theta_1$$

$$EX^2 = (EX)^2 + \text{var } X = \theta_1^2 + \theta_2$$

Emp. mom $E = \frac{1}{n} \sum X_i = \bar{X}$ $EX^2 = \frac{1}{n} \sum X_i^2$

$$\bar{X} = \theta_1$$

$$\sum x_i^2 = n(\theta_1^2 + \theta_2)$$

$$n\theta_2 = \sum x_i^2 - n\bar{x}^2$$

$$\stackrel{\text{check}}{=} \sum (x_i^2 - \bar{x}^2) = \sum (x_i - \bar{x})^2$$

$$= \sum x_i^2 - (\sum x_i)^2 / n$$

→ same estimates as MLE

Note that ~~for 1 parameter estimate~~
we need as many moments as # of param.

Exponential

Geometric distribution?

$$(Ex) f(x|p) = (1-p)^{x-1} p$$

$$L(p) = p^n (1-p)^{\sum x_i - n}$$

$$\ln L(p) = n \ln p + (\sum x_i - n) \ln(1-p) \quad 0 < p < 1$$

$$\frac{n}{p} - \frac{\sum x_i - n}{1-p} = 0$$

$$p = \frac{n}{\sum x_i} = \frac{1}{\bar{x}} \rightarrow \max$$

$$\boxed{\hat{p} = \frac{1}{\bar{x}}} = \frac{n}{\sum x_i} = \frac{\# \text{ successes}}{\# \text{ trials total}}$$

Geometric

$$\bar{x} = \frac{1}{p}$$

$$\hat{p} = \frac{1}{\bar{x}} = \frac{n}{\sum x_i}$$

$$E\left(\frac{1}{\bar{x}}\right) \neq \frac{1}{E(\bar{x})} = \frac{1}{p}$$

Biased

Check that MLE is the same.

$$\begin{aligned} \stackrel{n=1}{=} E\hat{p} &= E\left(\frac{1}{\bar{x}}\right) = \sum_k \frac{1}{k} P(X=k) = \sum_{k=1}^{\infty} \frac{1}{k} p(1-p)^{k-1} \\ &= p + \sum_{k=2}^{\infty} \frac{1}{k} p(1-p)^{k-1} > p \end{aligned}$$